△# 7.8 revision 1 and 3

CHAPTER 8 FURTHER APPLICATIONS OF INTEGRATION

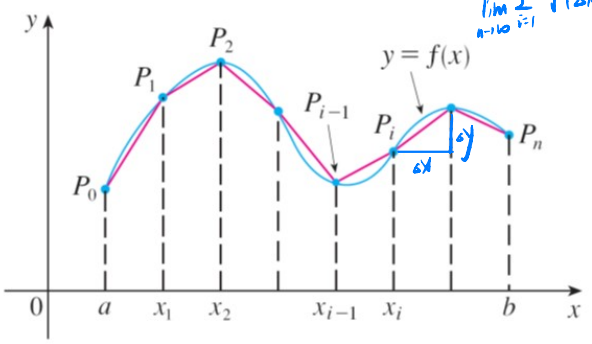
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8.1 Arc Length

- **Arc Length** 弧长

If f' is continuous on $[a, b]$, the arc length of the curve $y = f(x)$, $a \le x \le b$ is given by

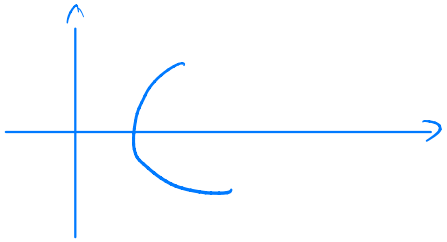
$$\int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$



$\lim_{n\to\infty} \sum_{i=1}^{n} \sqrt{(\Delta x_i)^2 + (\Delta y_i)^2} = \lim_{n\to\infty} \sum_{i=1}^{n} \sqrt{1 + \left(\frac{\Delta y_i}{\Delta x_i}\right)^2}$

$= \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$

$= \int_a^b \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy$



Example 8.1. Find the circumference of the circle $x^2 + y^2 = r^2$.

Example 8.2. Find the length of the line segment from $A(0, 1)$ to $B(5, 13)$.

Example 8.3. Find the length of the arc of the curve $y = x^{\frac{3}{2}}$ from the point $(1, 1)$ to the point $(4, 8)$.

$y = x^{\frac{3}{2}}$

$y' = \frac{3}{2} x^{\frac{1}{2}}$

Arc length $= \int_1^4 \sqrt{1 + (\frac{3}{2} x^{\frac{1}{2}})^2}\, dx$

$= \int_1^4 \sqrt{1 + \frac{9}{4} x}\, dx$

let $u = 1 + \frac{9}{4} x$; $\frac{du}{dx} = \frac{9}{4}$; $dx = \frac{4}{9}\, du$

when $x = 4$, $u = 10$

$x = 1$, $u = \frac{13}{4}$

$= \int^{10}$

If g' is continuous on $[c, d]$, the arc length of the curve $x = g(y)$, $c \le y \le d$ is given by

$$\int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy$$

Example 8.4. Find the length of the arc of the parabola $y^2 = x$ from $(0, 0)$ to $(1, 1)$.

$\frac{dx}{dy} = 2y$ Arc length $= \int_0^1 \sqrt{1 + (2y)^2}\, dy$

$= \int_0^1 \sqrt{1 + 4y^2}\, dy$

let $2y = \tan\theta$ when $\theta = 0$, $\theta = 0$

$y = \frac{\tan\theta}{2}$ $\theta = 1$, $\theta = \tan^{-1}(2)$

$dy = \frac{\sec^2\theta}{2}\, d\theta$

$= \int_0^{\tan^{-1}(2)} \sqrt{1 + \tan^2\theta} \cdot \frac{\sec^2\theta}{2}\, d\theta$

$= \int \frac{1}{2} \sec^3\theta\, d\theta$ $\int u\,dv = uv - \int v\,du$

let $u = \sec\theta$, $dv = \sec^2\theta\, d\theta$

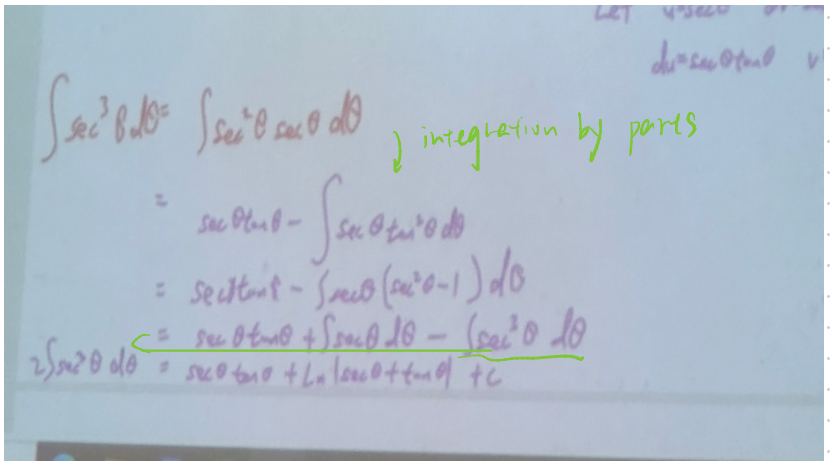
$du = \sec\theta\tan\theta\, d\theta$, $v = \tan\theta$

$= \left[\sec\theta\tan\theta\right]_0^{\tan^{-1}(2)} - \int \tan\theta \cdot \sec\theta\tan\theta\, d\theta$

$$\int \sec^3\theta\, d\theta = \int \sec^2\theta \sec\theta\, d\theta$$

$$= \sec\theta\tan\theta - \int \sec\theta\tan^2\theta\, d\theta$$

$$\int \sec\theta(\sec^2\theta - 1)$$



- **The Arc Length Function**

For the smooth curve $y = f(x)$, $a \le x \le b$, let $s(x)$ be the distance along C from the initial point $P_0(a, f(a))$ to the point $Q(x, f(x))$. Then

$$s(x) = \int_a^x \sqrt{1 + [f'(t)]^2}\, dt$$

Hence,

$$\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

Example 8.5. Find the arc length function of the curve $y = x^2 - \frac{1}{8}\ln x$ taking $P_0(1, 1)$ as the starting point.

$$s(x) = \int_1^x \sqrt{1 + \left(2t - \frac{1}{8t}\right)^2}\, dt$$

$$= \int_1^x \sqrt{1 + \left(4t^2 - \frac{1}{2} + \frac{1}{64t^2}\right)}\, dt$$

$$= \int_1^x \sqrt{4t^2 + \frac{1}{2} + \frac{1}{64t^2}}\, dt$$

$$= \int_1^x \sqrt{\left(2t + \frac{1}{8t}\right)^2}\, dt$$

$$= \int_1^x 2t + \frac{1}{8t}\, dt$$

$$= \left[t^2 + \frac{1}{8}\ln|t|\right]_1^x$$

$$= x^2 + \frac{1}{8}\ln|x| - 1$$

Exercises 8.1 9, 11, 15, 19, 33.

8.2 Area of Surface of Revolution

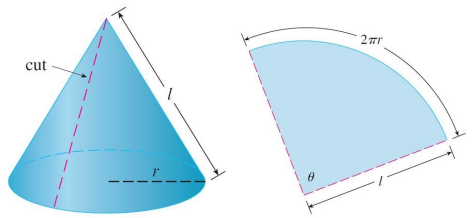
- **Surface Area**

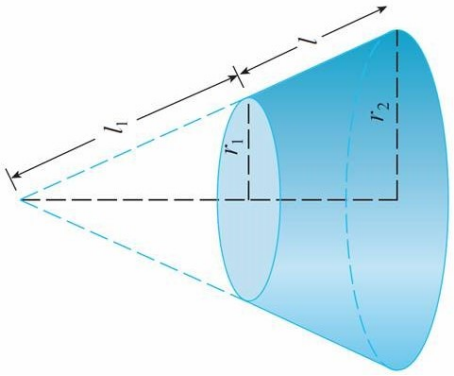
The area of the surface of revolution generated by rotating the curve $y = f(x)$, $a \le x \le b$ about the x-axis is

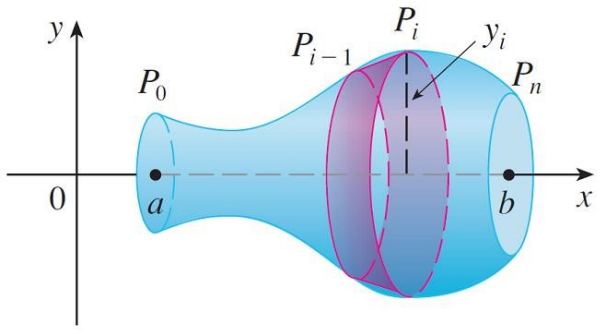
$$S = 2\pi \int_a^b y(x)ds = 2\pi \int_a^b y(x)\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx$$

The area of the surface of revolution generated by rotating the curve $y = f(x)$, $a \le x \le b$ about the y-axis is

$$S = 2\pi \int_a^b xds = 2\pi \int_a^b x\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx$$







Example 8.6. Find the area of the surface of revolution generated by revolving the curve $y = \sqrt{4 - x^2}$, $-1 \leq x \leq 1$ about the x-axis.

Example 8.7. The arc of the curve $y = x^2$ from $(1, 1)$ to $(2, 4)$ is rotated about the y-axis. Find the area of the resulting surface.

Exercises 8.2 7, 9, 13, 17, 31.